

2(a)(i)	<i>four from either</i> <i>idea of</i> making extracts of couch grass roots, of different ages / grown for different times / 14 days old / old(er) root(s) ; - grow barley (grains / young plants), supplied with (water containing) extract / has extract added ; - grow (another) set of barley (grains / young plants), (supplied with water) without extract ; or - grow couch grass for different times / to different ages / to 14 days / until older, and remove couch grass / cut off grass shoots ; - grow barley (grains / young plants) where couch grass has been previously grown and removed / where couch grass shoots had been cut off leaving roots ; - grow (another) set of barley (grains / young plants) on its own / where couch grass has not been grown ; then <i>ref. to</i> at least one standardised (environmental) condition ; - measure / record, length / (dry) mass, of barley roots ; <i>idea of</i> compare / analyse statistically, the length / (dry) mass / growth, of the barley roots ;	4 <i>I where barley / couch grass is grown, e.g. field, green house, plot, pot, paper in petri dish etc.</i> A extracts from separately sown couch grass or from couch grass from original experiment 2 A experiment 4 acts as / is, a control A <i>idea of</i> repeating experiment 2 but removing couch grass before barley is planted A <i>idea of</i> growing barley where only the roots are left A experiment 4 acts as / is, a control A e.g. same watering / temperature / light / humidity / time / nutrients / minerals I growth <i>unqualified</i> I measurement before investigation I compare growth of barley <i>unqualified</i> I chi squared test <i>must be clear that they have at least two treatments / values to compare</i>

Question	Answer	Mark	Additional Guidance
2(a)(ii)	<i>one from</i> 1 <i>idea that</i> established / older couch grass, is (better) competitor than barley for stated resources (light / minerals / water / space) / ora ; 2 <i>idea that</i> by the time barley is grown couch grass has depleted stated soil resources (light / minerals / water / space) ora ; 3 <i>idea of</i> older couch spreads a, disease / herbivore, to barley ; 4 <i>idea of</i> older couch produces a substance that inhibits / slows the germination of barley ; 5 <i>idea of</i> older couch grass changes the pH of the soil ;	1	A nutrients I nutrition I resources <i>unqualified</i> A nutrients I nutrition I resources <i>unqualified</i> A something that eats barley lives, in / on, older couch grass
2(a)(iii)	there is no significant difference between yield of barley grown with couch grass and, barley grown without couch grass ;	1	A there is no significant difference between yield of, barley grown with couch grass / experiment(s) 1 / 2 / 3, and, (the yield of) the control / experiment 4 A no significant decrease / increase in yield when couch grass is present compared to when couch grass is not present

Question	Answer	Mark	Additional Guidance										
2(b)(i)	correct calculation for both ground beetles ; <table><tr><td>ground beetles</td><td>20</td><td><u>0.181</u></td><td>45</td><td><u>0.012</u></td></tr></table> correct addition of both columns in table 2 ; <table><tr><td>total</td><td>47</td><td>0.300</td><td>414</td><td>0.188</td></tr></table> <i>correct values for both values of D</i> with pesticides $D=0.700$ and without pesticides $D=\underline{0.812}$;	ground beetles	20	<u>0.181</u>	45	<u>0.012</u>	total	47	0.300	414	0.188	3	<i>ecf for wrong values for ground beetles</i> A 0.7 / 0.70 ecf from wrong totals
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2(b)(ii)	two from 1 the use of pesticides reduces the numbers of all, the organisms / individuals / plants and animals ; 2 either the, biodiversity / species diversity, is reduced or idea that <i>D</i> / diversity index / biodiversity / species diversity, does not appear to be much affected / only changed by 0.112 ; 3 either use of processed data to describe percentage decrease in any one group or idea of beetles are less affected / have a much lower percentage decrease ; 4 bees (appear to have been) completely lost ; 5 idea that data collected is grouped, so cannot tell if any specific species has been lost ; 6 idea of reason for decline in, birds / small mammals , due to effect on food chain / non-specific nature of pesticides / herbicides ;	2	<i>must be in terms of effect of pesticides causing the decrease</i> A pesticides decrease the number of organisms I ora e.g. number of individuals in fields without pesticides is higher A species richness I diversity / <i>D</i> /, is reduced for either I diversity for or <table><tr><th>organism</th><th>percentage drop</th></tr><tr><td>dicot</td><td>95</td></tr><tr><td>monocot</td><td>88</td></tr><tr><td>beetles</td><td>56</td></tr><tr><td>butterflies and moths</td><td>89</td></tr><tr><td>bees</td><td>100</td></tr><tr><td>mammals</td><td>87</td></tr><tr><td>birds</td><td>94</td></tr><tr><td>total</td><td>89</td></tr></table>	organism	percentage drop	dicot	95	monocot	88	beetles	56	butterflies and moths	89	bees	100	mammals	87	birds	94	total	89
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